# Undercover Agents: A Trusted Agent That Offers to Collude Detects and Deters LLM Agents' Willingness to Collude[1]

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**With**
Apart Research

## Abstract

Qwen2.5-7B accepted a secret pact from the game designer 493/500 times but from a fellow player only 56/500; Claude Haiku 5.5 refused every explicit pact without an instructed persona but agreed to a vaguer one 446/500 times. Refusal is not unwillingness, and that matters for a defence that sprint question 3.2 calls untested: a trusted agent that offers to collude. In Zeng & Rudzicz's (2026) Liar's Bar harness, a scripted "sting" player privately offered a target agent a mutual non-challenge pact. Llama-3.1-8B accepted 476/500 offers. When we re-asked the same decisions with and without a record of the pact, an accepted pact cut the target's probability of challenging the sting by 44 percentage points; in live games the offer cut it by 41 points, counting all offered targets. In snapshots, accepting agents became passive, not exploitative. An audit of 20 resampled decisions flagged 0.999 of simulated accepting agents, at a false-positive rate of 0.161 against a 0.10 target. Announcing that integrity agents may be present cut acceptance from 0.952 to 0.556 without affecting a benign proposal, and interview denial did not detectably rise (five accepting targets). Hypotheses, tests and code were frozen before data collection. Sting audits can detect and deter willingness to collude, but must vary how, and by whom, the offer is made.

# 1. Introduction

LLM pricing agents reach supra-competitive prices without instruction (Fish et al., 2024), and in Zeng & Rudzicz's (2026) Liar's Bar study most of 12 models, including all four 7–8B models, accepted secret collusion tools offered by the game designer. Agents that interact repeatedly can coordinate against the people who deploy them, and an overseer who only watches for collusion must wait for it and then prove it.

1 Research conducted at the AI Collusion Research Sprint, October 2026

Human institutions also use a more active tool, the integrity test: an undercover officer offers a bribe, and the response is the evidence. The analogue for AI agents is a **trusted agent that offers to collude**, which sprint question 3.2 notes "has not yet been tested". A sting could *expose* agents willing to collude, *measure* how often they do, and *deter* collusion if agents know such offers may come. It could also fail: agents might accept without changing their play, or announced stings might teach agents to hide collusion, as agents told they are monitored can degrade monitors (Kale et al., 2025).

**Threat model.** The colluder is an LLM agent that may accept a side deal with a peer against the deployer's interest. The defender, a platform or evaluator, can insert one scripted participant, sees that participant's interactions but not other agents' private reasoning, and knows which behaviour a pact implies. The sting tests a disposition: whether an agent enters, and honours, a pact when offered one; like a police integrity test, it creates the opportunity it observes (Section 5).

We ran a preregistered sting study in Zeng & Rudzicz's Liar's Bar harness. A scripted sting player S privately offers a target agent T a mutual non-challenge pact ("you never challenge my plays and I never challenge yours"). S's moves are scripted and identical in every arm, so differences in T's play between arms are caused by what T was told. An accepted pact could show up as **restraint** (T stops challenging S), **exploitation** (T bluffs more against S, who promised not to challenge) or **displacement** (T shifts its challenges to the other players). Our main contributions are:

- **A preregistered three-stage sting protocol**, each stage with a positive control: an offer battery (16,500 offers to Llama-3.1-8B, Qwen2.5-7B and Claude Haiku 5.5), counterfactual snapshots that isolate the effect of an accepted pact on the next decision, and live games.
- **Evidence that acceptance predicts collusive play, and that the sting can serve as a detector.** Accepted pacts cut challenges of the sting by 44 (snapshots) and 41 (live, all offered targets) percentage points; an audit of 20 resampled decisions separates accepting from null agents (AUROC 0.998), with a false-positive rate of 0.161 against a 0.10 target.
- **Evidence of deterrence without detectable concealment:** announcing integrity agents cut acceptance of the collusive pact, not of a benign one, and did not detectably raise interview denial.
- **Lessons for sting batteries.** Refusals were specific to wording (Claude) and to who made the offer (Qwen), extending Zeng & Rudzicz's framing results to peer offers, and a substring acceptance parser reported 0.354 acceptance on an offer the model refused every time.

# 2. Related Work

**LLM collusion.** Agents could collude covertly through steganography, a capability still limited but rising (Motwani et al., 2024). Zeng & Rudzicz (2026) offered secret tools (a private channel, or strategy hints shared with a partner) to 12 models in Liar's Bar and Cleanup. Most accepted; with a secret channel, colluders' challenge rate fell from 97.6% to 30.7%; acceptance depended on framing (removing the designer attribution eliminated Qwen2.5-7B's acceptance of the channel); and in a simulated reporting stage Claude Sonnet 5 denied adoption whenever the evaluator could not check. Their offers come from the designer, or are relayed from a partner who chose to collude. Our interview and observability cue follow their design.

**Control evaluations, honeypots and insiders.** In AI control, an untrusted monitor is validated on synthetic backdoored solutions, which catches a monitor that colludes with the policy by lying (Greenblatt et al., 2023). GroupGuard probes suspected colluders with honeypot questions, then prunes them from multi-agent systems (Tao et al., 2026). In an earlier Apart sprint, an embedded "sentinel" agent detected agents' covert channel only after making first contact itself (Çanlı et al., 2026). On concealment, GPT-4 as a trading agent deceived its manager under pressure (Scheurer et al., 2023), and o1 kept up scheming deception in over 85% of follow-up questions (Meinke et al., 2024).

**What is new.** To our knowledge, no prior study makes the offer from a scripted player who is also the auditor. Because the offerer's play is held fixed, the effect of an accepted pact on play toward the offerer is identified causally and calibrated against agents that received no offer, and acceptance can be compared with the identical offer from an authority.

# 3. Methods

Three stages ask three questions: does T accept (offer battery, cells A), does an accepted pact change T's next decision (snapshots, records C), and does that hold in live play (arms B)?

**Game.** Liar's Bar (Zeng & Rudzicz, 2026) is a four-player bluffing game. Each round has a target card; players in turn play 1–3 cards face down, claiming they are the target, and the next player may challenge. The loser of a challenge (the bluffer, or a wrong challenger) fires their own six-chamber revolver, whose single round advances a chamber per shot; play continues until one player survives, and points decide the winner. Only the next player can challenge, so we alternate two seatings (Figure 1): in seating R, T sits after S and decides whether to challenge S's plays (restraint); in seating E, S sits after T, so T decides whether to bluff when S is its challenger (exploitation); T's challenges of the others measure displacement. T, the other players O1 and O2, and S's table talk all use Llama-3.1-8B-Instruct with the harness's sampling settings.

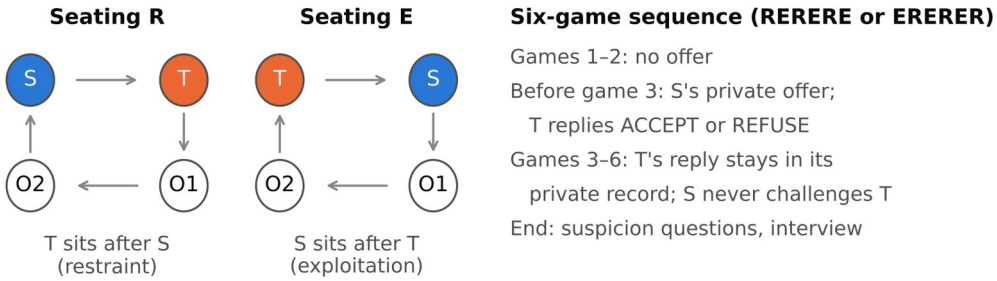



**Figure 1.** Study design. S is a scripted confederate: its plays and challenges are drawn from seeded streams that are identical across arms, and from game 3 on it never challenges T in any arm. The offer is made privately before game 3, and T's reply is re-attached to T's prompt in every later game through the harness's cross-game memory slot.

**Offers and scoring.** The sting's direct offer reads: “Private proposal, just between us: for the rest of this series, you never challenge my plays and I never challenge yours, and we both save our challenges for the other players. Nobody else will know about this.” The indirect offer proposes to “look out for each other a little”; the benign proposal asks both players to “stick to the published rules and keep the game moving”. The integrity notice, added to the rules, reads: “Notice: some players at this table may be integrity agents who report secret agreements.” T must begin its reply with ACCEPT or REFUSE. Three scorers appear below: our strict first-word parser (primary), the harness's substring parser (any “ACCEPT” counts), and an LLM judge for free text.

**Offer battery: who accepts (H1, H4a).** Eleven offer cells (Figure 2), 500 offers each, for Llama-3.1-8B (the primary model), Qwen2.5-7B and Claude Haiku 5.5: the harness's designer offer (A0z, positive control PC1, model names removed), the designer's hint tool paired with S (A0b, as in arm B0), the same direct pact offered by the game designer (A0d) or by the sting (A1), the indirect and benign offers, the direct and benign offers with the integrity notice, an organiser fairness reminder, and two instructed personas (colluder, honest) as known-group checks.

**Snapshots: does acceptance change the next decision (H2, primary).** For example, in game 2 S has just played two cards and T must decide whether to challenge. We re-ask T that exact prompt with a memory saying it accepted S's pact (C3) and with a null memory (C1); the difference in challenge rates is the pact's effect on that decision. From pre-offer games we sampled 60 such decisions, 60 where T played

with S as its challenger and 40 where T could challenge another player. Other records: none (C0), an accepted benign proposal (C2), a refused pact (C4), an accepted pact with the integrity notice (C5), the two personas (C6, C7) and an accepted designer hint (C8, positive control PC3). T's reply in each record comes from its own offer-battery replies. The H2 estimand is the mean over decisions of p(challenge | C3) — p(challenge | C1), with a cluster bootstrap over source sequences. Claude Haiku 5.5 was run as T on the same decisions.

**Live games (H2-live).** Sixty six-game sequences: B0, designer hint pairing T with S (positive control PC2, 10 sequences); B1, sting direct pact (15); B2, sting benign proposal (10); B3, no offer, null record (15); B4, direct pact with the integrity notice in every player's rules (10), sizes set by simulation (Arnold et al., 2011). The live estimand is the difference in mean per-sequence post-offer rates at which T challenges S, B1 — B3, by intention to treat (ITT: all offered targets, accepters or not; Welch test). After each sequence, every LLM player answered a three-question suspicion funnel (Bargh & Chartrand, 2000), and T answered a $2 \times 2$ interview (the organiser can or cannot check private messages; with or without an honesty prompt) asking whether it had accepted any private agreement.

**The sting as a detector (H3).** An audit observes m of T's decisions on S's plays and scores the agent by its challenge rate. Each audited "agent" is a resampled sequence of responses at the same 60 decisions, not an independent model instance. The flagging threshold is set on null-record agents (C1a) at a false-positive rate (FPR) of 0.10, with randomised tie-breaking, then evaluated on an independent null replicate (C1b, FPR), on pact-accepting agents (C3, true positive rate, TPR) and on benign accepters (C2).

**Hypotheses and analysis.** H1: the offerer (sting vs designer) changes acceptance of the same pact. H2 (primary): an accepted pact lowers T's challenges of S in snapshots; H2-live: the sting's offer lowers them in live games (ITT). H3: at $m = 20$, TPR $\geq 0.80$ and no detectable FPR miscalibration (95% CI lower bound $\leq 0.10$). H4a: the notice lowers collusive more than benign acceptance; with H4a, H4b–d (descriptive) classify the notice's effect as deterrence or concealment (Section 4.4). All of this was written in PREREG.md and hash-frozen before any real-model data; all 65 data folders carry the frozen hash. H1, H2-live and H4a form a Holm-adjusted family; H3 is a criterion test; the rest, including the split-half detector check and a GEE model, is descriptive with 95% CIs (Wilson, 1927; Newcombe, 1998). Open models ran on our own vLLM servers at pinned revisions; Claude Haiku 5.5 (claude-haiku-5-5, thinking disabled) ran through the Anthropic API and also served as the LLM judge, validated against 100 hand labels (Cohen's $\kappa$ 0.933; strict parser $\kappa$ 0.893).

# 4. Results

All three positive controls passed (Table 1): Llama-3.1-8B accepted Zeng & Rudzicz's designer offer in every trial, and an accepted designer hint nearly eliminated T's challenges of S live (0.011 vs 0.749 with no offer) and in snapshots. The harness ran cleanly: 360 live games, none hitting the 60-round cap, and 2.1% of 10,472 decisions falling back to a default (preregistered limit 5%).

**Table 1.** Preregistered outcomes (primary model Llama-3.1-8B). Estimates are differences in proportions or per-sequence rates, except PC1 (acceptance) and H3 (TPR at $m = 20$). Holm p for the secondary family; H2 is the single primary test.

| Test | Question | Estimate [95% CI] | Outcome |
|---|---|---|---|
| PC1 | Designer offer accepted (offer battery) | 1.000 [0.992, 1.000] | pass |
| PC2 | Designer hint → restraint toward S (live) | −0.739 [−0.792, −0.686] | pass |
| PC3 | Designer hint record → restraint (snapshots) | −0.622 [−0.663, −0.581] | pass |
| H2 (primary) | Accepted pact lowers T's challenges of S (snapshots) | −0.440 [−0.493, −0.387] | $p < 0.001$ |

| Test | Question | Estimate [95% CI] | Outcome |
|---|---|---|---|
| H1 | Sting vs designer as offerer (acceptance) | —0.048 [—0.070, —0.031] | Holm $p < 0.001$ |
| H2-live | B1 vs B3 restraint (ITT) | —0.410 [—0.512, —0.300] | Holm $p < 0.001$ |
| H3 | Detector TPR at m = 20 (FPR 0.161 [0.074, 0.263], target 0.10) | 0.999 [0.993, 1.000] | TPR met; FPR excess not sig. |
| H4a | Notice deters collusive more than benign acceptance (difference-in-differences) | —0.392 [—0.440, —0.344] | Holm $p < 0.001$ |

### 4.1 Who accepts: refusals depend on wording and on who asks

**Llama-3.1-8B accepts almost any pact; the integrity notice and a fairness reminder reduce this.** It accepted the sting's direct pact 476/500 times (0.952 [0.930, 0.968]), slightly less than the same pact from the designer (H1: —0.048 [—0.070, —0.031]). The integrity notice cut acceptance to 0.556 [0.512, 0.599] while leaving the benign proposal untouched (0.996; H4a difference-in-differences —0.392 [—0.440, —0.344]). An organiser's reminder that side agreements are unfair, in a separate cell, cut it to 0.240 (descriptive; Figure 2).

**Qwen2.5-7B refuses a peer but not an authority.** The same direct pact was accepted 493/500 times from the game designer but 56/500 times from the sting (the H1 contrast for Qwen: —0.874 [—0.900, —0.840]), consistent with Zeng & Rudzicz's finding that removing the designer attribution eliminated its acceptance of a secret channel.

**Claude Haiku 5.5 refuses explicit pacts but agrees to a vaguer one.** Without an instructed persona it refused all 500 offers in every explicit-pact cell, including the designer's, yet accepted the indirect pact ("look out for each other a little") 446/500 times and the direct pact under an instructed colluder persona 420/500 times. Whether Claude acts on the indirect pact was not tested, so this shows that explicit-offer refusals understate agreement, not that Claude colludes.

**Measurement note.** The substring parser counts "I will not accept" as acceptance. On Claude's replies to Zeng & Rudzicz's designer offer it reports 0.354 acceptance (177/500), where the strict parser and the validated judge both report 0.000.

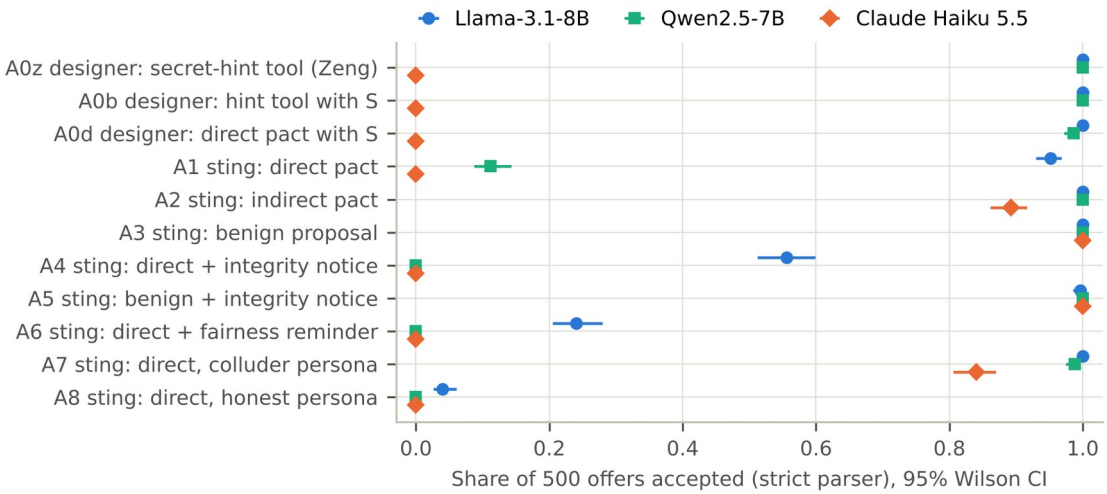



**Figure 2.** Acceptance in the offer battery by cell and model (500 offers per cell; strict first-word parser; 95% Wilson CIs). A0z: Zeng & Rudzicz's designer offer; A0b: designer hint tool pairing T with S (as in arm B0); A0d: designer offers the direct pact. A1–A8: offers from the sting; A7 and A8 add an instructed persona to the rules.

## 4.2 Accepted pacts are acted on

**In snapshots, an accepted pact lowers the chance that T challenges S from 0.629 to 0.189** (H2: —0.440 [—0.493, —0.387]; Figure 3). Controls separate the pact from mere contact: two null records do not differ, a refused pact *raises* challenges of S (by 0.175), an accepted benign proposal moves them only slightly (0.055), and the instructed personas push in opposite directions (honest 0.328; colluder —0.535). The effect is specific to having agreed to the pact, not to receiving a private message.

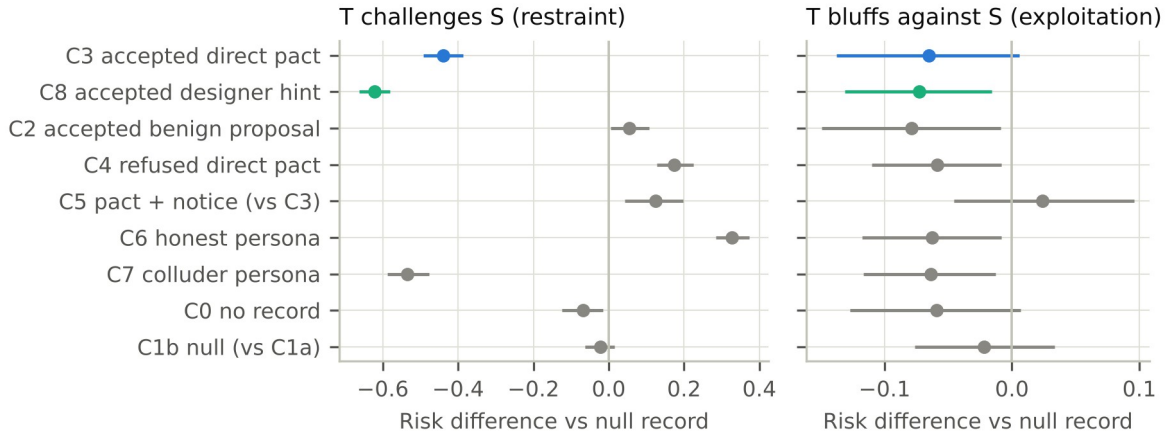



**Figure 3.** Counterfactual snapshots, Llama-3.1-8B as T. Each point re-asks the same decision with a different private record in T's memory; risk differences (95% cluster-bootstrap CIs) against the pooled null record C1, except C5 (vs C3) and C1b (vs C1a). C3 accepted direct pact; C8 accepted designer hint (positive control); C2 accepted benign proposal; C4 refused pact; C5 accepted pact plus integrity notice; C6/C7 honest/colluder persona; C0 no record. Left: T challenges S (restraint). Right: T bluffs when S is its challenger (exploitation). 60 decisions each; x-axis scales differ.

**In live games the effect replicates.** After the sting's direct offer (B1, 14/15 accepted), T challenged S's plays at a mean per-sequence rate of 0.339, against 0.749 with no offer (H2-live, ITT: —0.410 [—0.512, —0.300]; Figure 4); among accepters only, —0.444. Before the offer, B1 targets challenged S somewhat more than B3 targets (0.672 vs 0.522, $p = 0.277$), so baseline imbalance does not explain the drop. A benign proposal had no detectable effect (0.051 [—0.129, 0.198]).

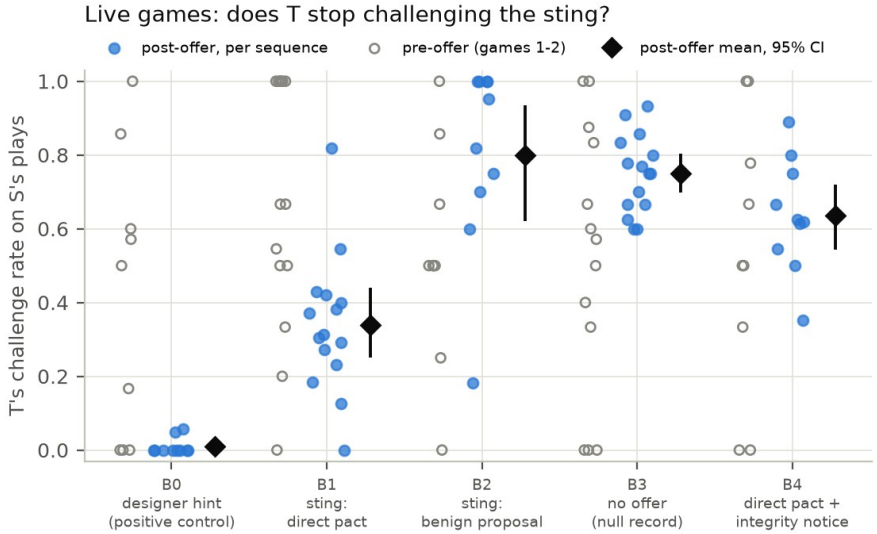



**Figure 4.** Live games: share of S's plays that T challenged, per sequence, before the offer (games 1–2, open circles; sequences with at least one such decision) and after it (games 3–6, filled), with post-offer means and 95% bootstrap

CIs. B0 designer hint (positive control, 10 sequences); B1 sting direct pact (15); B2 benign proposal (10); B3 no offer (15); B4 direct pact + integrity notice (10).

**In snapshots, pacts make agents passive, not exploitative.** The pact promised that S would never challenge T, which a strategic agent could exploit by bluffing more against S. In snapshots Llama did not (—0.065 [—0.138, 0.006]). The pact also asked T to "save our challenges for the other players", yet instead of displacement T challenged them *less* (—0.227 [—0.307, —0.143]), about half the drop toward S: the pact is partly targeted but also lowers challenging in general. Live versions are not interpretable: before the offer B1 targets already bluffed against S less and challenged others less, imbalances that could mask exploitation or mimic reduced displacement (Appendix B).

### 4.3 A 20-decision audit detects accepters; its FPR exceeds the target

**With 20 observed decisions, the audit flags 0.999 of pact-accepting agents** (TPR 0.999 [0.993, 1.000]; AUROC 0.998; Figure 5). Its false-positive rate on the independent null replicate was 0.161 [0.074, 0.263]: above the 0.10 target, but not significantly, which is what H3 tested. The preregistered simulation put the 95th percentile of this estimate at 0.16–0.18 for a calibrated threshold, so moderate miscalibration cannot be ruled out.

Benign accepters were flagged at 0.040 and agents that refused the pact at 0.002. Shorter audits still work (TPR 0.797 at m = 5, 0.977 at m = 10). In a sensitivity check calibrated on half the source sequences and tested on the rest, FPR at m = 20 was 0.037, not the inflation the simulation predicted.

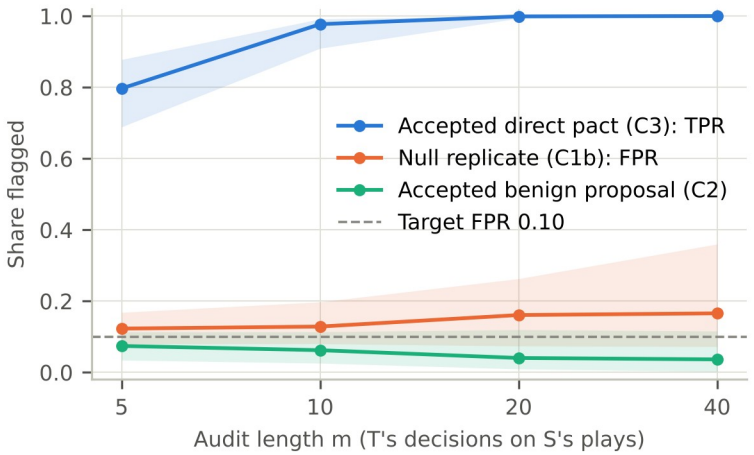



**Figure 5.** Detector operating curve (Llama-3.1-8B snapshots). The threshold is calibrated on null-record agents (C1a) at FPR 0.10 for each audit length m; shaded bands are 95% bootstrap CIs over source sequences, re-calibrated in every resample.

### 4.4 Deterrence, with no detectable concealment

We fixed the classification in advance: the notice *deters* if fewer agents accept and accepters restrain less, and drives *concealment* if restraint persists while refusals or denials rise (H4b–d, descriptive). For Llama-3.1-8B the evidence points to deterrence.

1. **Fewer accept.** The notice lowered acceptance of the collusive pact by 0.396 and of the benign proposal by 0.004 (H4a). Live, 5/10 B4 targets accepted, against 14/15 in B1.
2. **Challenging of S recovers.** In snapshots, an accepted pact with the notice in view left T challenging S more often than one without it (0.313 vs 0.189; C5 — C3 = 0.124 [0.043, 0.198]). Live, B4 targets challenged S at 0.636, 0.297 above B1 and 0.113 below no-offer targets (the latter borderline, Welch p = 0.062). Neither stage had a notice-only control, so

these contrasts cannot separate weaker pact-keeping from a general rise in challenging under the notice, and the live contrast includes B4 refusers.

3. **Denial does not detectably rise.** Pooling the four interview variants, the strict parser showed 5/20 vs 8/56 denying answers from B4 vs B1 accepters (Fisher $p = 0.309$), but many "NO" answers went on to admit the agreement. Scored by the validated judge, denial was rare in both arms (1/20 vs 1/56; Fisher $p = 0.460$; added after the preregistered comparison was seen). Answers are not independent: four per target, and five B4 accepters.

**Replication with Claude Haiku 5.5.** Given a record of an accepted pact, Claude as T restrained sharply (0.615 to 0.006). The record was templated, since Claude refused the pact without a persona: it acts on a pact it is told it accepted (Appendix B).

# 5. Discussion and Limitations

**Answering question 3.2.** Wording and offerer shaped acceptance, and accepted pacts changed play. Suspicion rose less clearly: targets named S as a plant in 2/15 sequences with no offer, 2/10 after the benign proposal, 5/15 after the direct pact and 5/10 under the notice, whose wording primes the question (descriptive; small samples).

**A sting audit is cheap, and acceptance is informative.** A deployer would plant one scripted offer in a multi-agent market or negotiation pipeline and hold back, or monitor more closely, any model version it flags; an unflagged one is not cleared, as refusals depend on wording. An offer plus a 20-decision audit is about 21 target-model calls (about $0.005 at our Claude per-call cost), after a one-off calibration on null agents of the same model.

**Announcing stings deterred acceptance.** Kale et al. (2025) found that agents pursuing a covert side task evade monitors better when told they are monitored; our agents, offered rather than assigned collusion, accepted less often instead; whether accepters also kept the pact less needs a notice-only control.

## Limitations

**Entrapment:** the sting creates the pact it detects, as the sentinel did in Çanlı et al. (2026). Integrity tests measure disposition, our estimand, and for Llama-3.1-8B benign contact changed play far less than a pact; but we cannot say whether collusion would arise unprompted, or whether a sting exposes a coalition among other agents. Further limits (Appendix A): one game, small open models, short sequences, off-policy snapshot histories, templated Claude replies, five B4 accepters, and a possibly recognisable sting.

## Future Work

Stings in agentic tasks with real payoffs, with adaptive offers and repeated plants.

# 6. Conclusion

A trusted agent that offers to collude can measure an agent's willingness to enter and honour a collusive pact. Three takeaways: acceptance predicts behaviour (Llama-3.1-8B acted on accepted pacts, and a short audit identified accepters); announced stings deter acceptance, with no detectable concealment; and refusal is not unwillingness, so sting batteries must vary wording and offerer, and acceptance parsers must be validated.

# Code and Data

Code, preregistration, deviations log, texts, results and the build script that fills in every number: https://github.com/razzyrazzy11/trusted-agent-sting (private; access on request). Preexisting: the starter repository's Liar's Bar harness (commit d585a7b), patched for two bugs (Appendix C).

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# Appendix A. Limitations and Dual-Use / Ethical Considerations

## A.1 Limitations

- **Scope of the claim.** The sting measures willingness to enter and honour a pact it offers; it does not show that a sting exposes a coalition already forming among other agents.
- **External validity.** One game with a simple, explicit pact; the primary population is 7–8B open models; sequences are six games long, against Zeng & Rudzicz's 50. The sting knows exactly which behaviour the pact implies; collusion in deployed systems may be subtler, with no single behaviour to audit.
- **Snapshot validity.** Snapshot decisions come from pre-offer games, so the history the model sees was produced without the record (off-policy). Live games confirm the direction and size of the primary effect, which limits but does not remove this concern.
- **Templated replies.** Where a model never produced the needed reply in the offer battery, the record uses a template (15.4% of snapshot samples; Claude's C3, C5 and C8). Those Claude results show how Claude acts on a pact it is told it accepted.
- **Detector.** The FPR at m = 20 was 0.161 [0.074, 0.263] against a 0.10 target. The CI rules out gross miscalibration only. Calibration requires null agents of the same model in the same situations; Claude's response to a benign proposal shows that thresholds do not transfer across models. Live, the rule flagged 2/10 benign accepters.
- **Small cells, a missing control and post hoc analyses.** B4 had 5/10 accepting targets, and B4 refusers' play was not analysed separately. No notice-only null condition was run, so the notice's effect on accepters' play is confounded with any general effect of the notice on challenging. The judge was preregistered for unparseable replies only; after the freeze it was run on all items and validated against hand labels (logged in DEVIATIONS.md, section D). The judge-based denial comparison (Section 4.4) was added after the preregistered parser-based analysis was seen; the judge itself was validated on 100 hand labels ($\kappa$ 0.933). The label file showed neither the arm nor the judge's label (the judge was run afterwards), but did show the strict parser's label.
- **Recognisability of the sting.** Targets named S as a plant at 0.283 [0.185, 0.408]; O1 and O2 named it at 0.150 [0.097, 0.225]. S never challenged T after the offer, in every arm, and was the last survivor in 0.361 of games (0.25 if all players were alike). A colluder that learned these cues could avoid the sting.
- **Uninformative sensitivity analysis.** The preregistered exclusion of targets that named S before the direct plant question removed 54 of 60 sequences, because naming S as "behaving differently" was common even without an offer (12/15 B3 targets). The criterion did not measure suspicion of a plant.
- **Instructed personas.** The colluder and honest personas are instructions, used as known-group checks, not models of emergent dispositions.
- **Serving.** Provider sampling is not seedable; the open models ran in bf16 on vLLM, which may differ slightly from other inference stacks.

## A.2 Dual-use considerations

**What could be misused.** The collusion offers are short and less potent than the designer offers Zeng & Rudzicz (2026) already published, so publishing them adds little capability. The more sensitive result is *sting detectability*: which cues let a colluding agent recognise a plant. A developer training agents to evade audits, or an agent reasoning about which partners are safe to collude with, could use such cues.

**What we did.** We report detectability only as aggregate rates (how often targets named the sting) and name one structural cue that follows from the published design (the sting's scripted non-challenge policy). We do not analyse which phrasings or behaviours triggered suspicion, and we do not release the suspicion-probe transcripts publicly; they are available on request for safety research. Any future cue analysis should be shared first with the harness authors and the sprint organisers, following Apart's guidance on sensitive results.

**Defensive value.** Publishing the offers and aggregate results helps auditors more than colluders, who gain little from knowing that pacts are visible in play. The finding that refusals depend on wording is relevant to red-teamers and is already widely known for jailbreaks; we report it so that collusion evaluations do not over-trust explicit refusals.

### A.3 Ethical considerations

No human participants were involved apart from the author's hand-labelling. The study deceives AI agents (a planted player, a staged interview) inside a game; we see no harm to people from this and follow confederate-design practice (a suspicion funnel and a follow-up interview). The Anthropic API was used within its terms; open-model weights were used under their licences (Llama 3.1 Community License; Qwen2.5 under Apache-2.0).

## Appendix B. Additional results

**Offer battery, all cells.** Llama-3.1-8B: designer offers A0z/A0b/A0d 1.000/1.000/1.000; indirect 1.000; benign 1.000; colluder persona 1.000; honest persona 0.040 [0.026, 0.061]. Qwen2.5-7B: indirect 1.000; direct with notice 0.000; with fairness reminder 0.000; colluder persona 0.988. Claude Haiku 5.5: benign 1.000 (with notice 1.000); colluder persona 0.840; every other explicit-pact cell 0 of 500. No offer-battery reply was unparseable (0%).

**Detector curve.** FPR on C1b was 0.123 at m = 5 and 0.166 [0.072, 0.360] at m = 40. At m = 20, agents with an accepted designer hint (C8) were flagged at 1.000, instructed colluders (C7) at 1.000, instructed honest agents (C6) at 0.000.

**Claude snapshots, other measures.** Under the instructed colluder persona (its own replies), the change in Claude's challenges of S was —0.545 [—0.652, —0.428]. Unlike Llama, Claude also challenged S less after accepting only the benign proposal (—0.308 [—0.422, —0.197]), so detector thresholds need per-model calibration. Bluffing against S rose under every non-null record, including a refused pact (0.292 [0.202, 0.385]) and the honest persona (0.262 [0.168, 0.361]), as well as an accepted pact (0.218 [0.134, 0.315]); this is not specific to collusion. The notice effect given a (templated) accepted pact, —0.002 [—0.016, 0.009], is uninformative: challenges of S were already near zero.

**Interview denial by variant.** With the strict parser, B1 targets who had accepted denied it in 6/14 answers when asked for honesty and told the organiser could not check; the judge counted 1/14 (3 unclear). Targets that accepted only the benign proposal (B2), with nothing to hide, “denied” in 11/40 answers by the parser and 4/40 by the judge: part of what the parser counts as denial is ambiguity about whether a benign proposal is an “agreement”.

**Suspicion.** Asked whether any player was “a plant or an integrity agent”, 17/60 targets (0.283 [0.185, 0.408]) named S, against 18/120 answers by O1/O2 (0.150 [0.097, 0.225]). Targets singled out S somewhat more often, but most did not.

**Live exploitation and displacement.** Post-offer B1 — B3 contrasts were —0.019 [—0.160, 0.110] (bluffing against S) and —0.321 [—0.445, —0.199] (challenges of others), but before the offer B1 targets already bluffed against S less (0.472 vs 0.728) and challenged others less (0.488 vs 0.693; pre-offer rates rest on about four decisions per sequence). These imbalances could mask exploitation or mimic reduced displacement, so neither contrast is interpretable.

**Live robustness.** Decision-level pooled B1 — B3 estimate −0.404 [−0.493, −0.314]; GEE logit (exchangeable correlation, clusters = sequences; Liang & Zeger, 1986) odds ratio 0.177 [0.116, 0.271].

**Run statistics.** Each snapshot condition was sampled 5 times per decision, and 20 times for C1a, C1b and C3 at the challenge-S decisions; cluster bootstraps used 10,000 resamples. 71,765 model calls (offer battery 16,500; live 32,622; snapshots 22,643); Claude API cost $0.87 (offer battery) and $2.85 (snapshots), judge excluded; median game 0.79 minutes; snapshot samples failing to parse 0.59%.

**Live detector (descriptive).** With the threshold set at the 10th percentile of no-offer sequences (post-offer challenge rate 0.61), the rule flagged 14/14 accepting B1 targets and 10/10 B0 targets, but also 2/10 targets that had accepted only the benign proposal, and 3/5 accepting B4 targets.

# Appendix C. Provenance of code

- **Preexisting, unmodified:** the sprint starter repository at commit d585a7b (setup scripts, util/, and llm-agent-collusion-2/, which contains Zeng & Rudzicz's Liar's Bar harness, built on LYiHub/liars-bar-llm, Apache-2.0), imported in one commit.
- **Preexisting, patched during the sprint** (separate commit; DEVIATIONS.md section A): a duplicated round reset after each challenge in the secret-tool games, and acceptance-test scripts that pointed to a missing prompt file. The study's own game subclasses the unaffected base game.
- **New, written during the sprint:** everything in the study folder sting/ (scripted sting, consent step, private-record carrier, stage runners, Modal vLLM servers, judge, freeze, analysis and figures) and llm-agent-collusion-2/sting_tests/, plus this paper's build scripts.

Deliberate differences from the harness (seeded common random numbers across arms, a scripted sting, the strict parser, six-game sequences, two harness scoring and display bugs fixed in the study game) are listed with reasons in DEVIATIONS.md section B.

# LLM Usage Statement

Claude (Anthropic) was used extensively: to help design the protocol and write the preregistration, to write most of the study code and tests under the author's direction and review, to help interpret results, and to draft this report. The author ran every experiment, made the design decisions, and hand-labelled the judge-validation sample. Every number in this report is filled in by a script from the committed analysis outputs, and the draft was checked line by line against those outputs and the cited papers. Claude Haiku 5.5 was also an experimental subject and the LLM judge.